

EXAM CRACKER & PYQ ANALYSIS

BT-101: Engineering Chemistry (Grading System)

Comprehensive Blueprint & Question Bank (2020 — 2025) | Structured Unit-Wise & Topic-Wise

Strategic Insights & Usage Guide

This blueprint has been generated by analyzing 10 consecutive previous year question papers of RGPV B.Tech (I & II Semester) from **June 2020 to December 2025**. Questions are strictly cataloged by frequency and RGPV syllabus units to optimize your revision. Note that although the prompt referenced BEEE BT(104), all provided official PYQ materials belong to **BT-101 Engineering Chemistry**, which is the subject fully decoded below.

UNIT 1: WATER ANALYSIS, TREATMENT & BOILER TECHNOLOGY

Topic 1.1: Hardness of Water & EDTA Method

Key Relations & Formulae:

- Hardness equivalents in terms of CaCO_3 = *Weight of hardness causing salt* $\times$ (*100 / Equivalent weight of salt*)
- Inter-relationship: $1 \text{ ppm} = 1 \text{ mg/L} = 0.07^\circ\text{Clark} = 0.1^\circ\text{French}$

Question Text	Classification	Occurrences / Years	Type
Define Hardness of water. Explain the various units of hardness and their inter-relationship. Why is EDTA method preferred over other methods for hardness determination?	MOST IMPORTANT	5 Times Dec 2025, Jun 2024, Jun 2022, Dec 2020, Jun 2020	Theory
Explain the principle of EDTA titration. Describe the estimation of total hardness of water by complexometric method.	FREQUENTLY ASKED	3 Times Dec 2025, Nov 2022, Dec 2020	Theory & Principle
Differentiate between hard water and soft water. What is temporary and permanent hardness of water and how are they differentiated? Give examples.	FREQUENTLY ASKED	3 Times Dec 2024, Jun 2024, Jun 2020	Theory
Numerical Pattern A (EDTA): 100 ml of a water sample required 18 ml of standard EDTA solution. 20 ml of standard CaCO_3 (1.5g/L) solution required 25 ml of EDTA solution. 100 ml of sample after boiling required 12 ml of EDTA. Calculate temporary and permanent hardness.	FREQUENTLY ASKED	2 Times Dec 2023, Dec 2025 (similar)	Numerical

Question Text	Classification	Occurrences / Years	Type
Numerical Pattern B (Constituents): Calculate the total hardness (in ppm) of a water sample containing: $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$, $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$, and $\text{CaSO}_4 = 13.6 \text{ mg/L}$.	MOST IMPORTANT	3 Times Dec 2025, Jun 2025, Nov 2022	Numerical

Topic 1.2: Boiler Troubles & Industrial Applications

Question Text	Classification	Occurrences / Years	Type
What are boiler troubles? Explain the formation, composition, prevention, and disadvantages of Scales and Sludges in boilers. How do they differ?	MOST IMPORTANT	6 Times Jun 2025, Dec 2024, Jun 2024, Dec 2023, Jun 2023, Jun 2020	Theory & Prevention
Write short notes on Caustic Embrittlement and Boiler Corrosion . Explain their causes and minimization techniques.	MOST IMPORTANT	5 Times Jun 2025, Dec 2023, Jun 2023, Nov 2022, Dec 2020	Theory
Discuss the following terms: (i) Blow-down process, (ii) Carry-over (Priming and Foaming).	MODERATE IMPORTANCE	2 Times Jun 2024, Dec 2024	Short Answer

Topic 1.3: External Water Treatment Processes (Lime-Soda, Zeolite, Ion-Exchange, RO)

Question Text	Classification	Occurrences / Years	Type
Describe the Ion-Exchange (Deionization) process for water softening with a neat labeled diagram and chemical reactions. Why is it superior for high-pressure boilers?	MOST IMPORTANT	6 Times Dec 2025, Jun 2025, Dec 2024, Jun 2024, Nov 2022, Jun 2020	Theory + Diagram
Discuss the Zeolite (Permutit) process for water softening. Compare it with the Ion-Exchange process.	FREQUENTLY ASKED	4 Times Dec 2025, Dec 2023, Jun 2023, Nov 2022	Theory
Explain the chemical reactions involved in the Lime-Soda process (Hot/Cold). Why is a small amount of coagulant (like Alum) added? Calculate Lime/Soda requirements.	MOST IMPORTANT	5 Times Dec 2025, Dec 2023, Jun 2023, Jun 2022, Dec 2020	Theory + Numerical

Question Text	Classification	Occurrences / Years	Type
Numerical Pattern (Lime-Soda): Calculate requirement of Lime (80% pure) and Soda (90% pure) for treatment of 1,00,000 liters of water containing standard salts.	FREQUENTLY ASKED	3 Times Dec 2025, Jun 2025, Jun 2023	Numerical
What is Desalination? Describe the Reverse Osmosis (RO) process with a neat labeled diagram and state its industrial applications.	MODERATE IMPORTANCE	1 Time Dec 2025	Theory + Diagram

Topic 1.4: Water Alkalinity & Disinfection

Question Text	Classification	Occurrences / Years	Type
Discuss the method for determination of Alkalinity in a given water sample. Calculate alkalinity types (OH^- , CO_3^{2-} , HCO_3^-) when 100 ml sample requires phenolphthalein and methyl orange endpoints.	FREQUENTLY ASKED	4 Times Jun 2025, Jun 2022, Dec 2020, Jun 2020	Numerical & Theory
How is portable water disinfected by Ozonation ?	RARE BUT POSSIBLE	1 Time Nov 2022	Short Answer



Unit 1 Expected Questions:

- Numerical on Lime-Soda requirements or EDTA hardness estimation (Guaranteed 7/14 marks).
- Comprehensive long answer comparing Zeolite vs Ion-Exchange with a clean schematic.
- Distinction table for Scales vs Sludges and steps to eliminate them.

UNIT 2: LUBRICANTS & POLYMERS

Topic 2.1: Lubricants – Mechanisms & Core Properties

Question Text	Classification	Occurrences / Years	Type
Define Lubrication . Explain the mechanisms of (i) Thick film/Fluid film, (ii) Thin film/Boundary, and (iii) Extreme Pressure Lubrication. Which type is used for heavy-duty machinery?	MOST IMPORTANT	5 Times Dec 2025, Dec 2024, Jun 2024, Jun 2023, Nov 2022	Theory
Discuss the importance of Viscosity Index (VI) . How does it affect engine oil performance under varying temperatures? Explain its calculation method.	MOST IMPORTANT	6 Times Dec 2025, Jun 2025, Jun 2024, Dec 2023, Jun 2022, Dec 2020	Theory + Numerical
Explain the significance of the following properties of lubricants: (i) Flash & Fire point, (ii) Cloud & Pour point, (iii) Aniline point, (iv) Saponification/Neutralization value, (v) Steam Emulsion Number.	MOST IMPORTANT	7 Times Jun 2025, Dec 2024, Jun 2024, Dec 2023, Jun 2023, Dec 2020, Jun 2020	Properties Theory
Numerical Pattern (Viscosity Index): Calculate VI of an oil sample whose Saybolt Universal Viscosity matches standard Gulf oil and Pennsylvanian oil at 210°F and 100°F .	MODERATE IMPORTANCE	1 Time Jun 2025	Numerical

Topic 2.2: Polymers – Mechanisms & Industrial Resins

Question Text	Classification	Occurrences / Years	Type
Differentiate between Chain-growth (Addition) and Step-growth (Condensation) polymerization. Explain Free Radical and Ionic mechanism with examples.	MOST IMPORTANT	6 Times Dec 2025, Jun 2025, Jun 2024, Jun 2023, Nov 2022, Dec 2020	Mechanism Theory
Differentiate between Thermoplastic and Thermosetting resins with examples.	FREQUENTLY ASKED	3 Times Jun 2025, Nov 2022, Dec 2024	Distinction

Question Text	Classification	Occurrences / Years	Type
Give the preparation/synthesis, properties, and engineering applications of: (i) Bakelite , (ii) PVC , (iii) PMMA , (iv) Nylon 6,6 , (v) Urea-Formaldehyde.	MOST IMPORTANT	7 Times Dec 2025, Jun 2025, Dec 2024, Dec 2023, Jun 2022, Nov 2022, Dec 2020	Synthesis & App
Write short notes on: (i) Vulcanization of Rubber, (ii) Synthetic Rubber (Buna-S, Neoprene), (iii) Biodegradable polymers, (iv) Conducting Polymers.	FREQUENTLY ASKED	5 Times Dec 2025, Jun 2024, Dec 2023, Jun 2023, Jun 2022	Short Notes

UNIT 3: CORROSION & PHASE EQUILIBRIUM

Topic 3.1: Corrosion Theories, Mechanisms & Control

Question Text	Classification	Occurrences / Years	Type
Define Corrosion. Explain the Electrochemical (Wet) theory of corrosion with reference to rusting of iron.	MOST IMPORTANT	4 Times Jun 2025, Jun 2023, Dec 2020, Jun 2020	Mechanism Theory
Explain the Differential Aeration Theory of corrosion with the help of Pitting corrosion and Waterline corrosion examples. Why do iron nails rust faster in saline water?	MOST IMPORTANT	4 Times Dec 2025, Dec 2024, Jun 2024, Jun 2022	Theory Case Study
What is Oxidation (Dry) Corrosion ? Describe its mechanism and types of oxide films formed (Pilling-Bedworth rule).	FREQUENTLY ASKED	3 Times Dec 2024, Jun 2024, Jun 2022	Theory
Discuss corrosion prevention methods. Differentiate between Sacrificial Anodic Protection and Impressed Current Cathodic Protection . Explain Galvanization.	FREQUENTLY ASKED	3 Times Dec 2025, Nov 2022, Jun 2020	Prevention
Numerical Pattern: How much rust ($Fe_2O_3 \cdot 3H_2O$, Mol Wt = 214g) will be formed when 100 kg of iron have completely rusted away?	RARE BUT POSSIBLE	1 Time Jun 2025	Numerical

Topic 3.2: Phase Rule & Phase Equilibrium

Gibbs Phase Rule: $F = C - P + 2$ (For standard systems) | $F = C - P + 1$ (For condensed/two-component systems)

Question Text	Classification	Occurrences / Years	Type
Define and explain the terms involved in Phase Rule: Phase (P) , Component (C) , and Degree of Freedom (F) .	FREQUENTLY ASKED	3 Times Jun 2025, Jun 2022, Dec 2020	Definitions
Draw a neat labeled Phase Diagram of Water System (One-component). Explain the curves, areas, triple point, and metastable curve (OA') inside it.	MOST IMPORTANT	5 Times Jun 2025, Jun 2024, Jun 2023, Jun 2022, Dec 2020	Diagram & Theory

Question Text	Classification	Occurrences / Years	Type
Explain a two-component system by taking the Copper-Silver (Cu-Ag) system or Eutectic system as an example. State the significance of the Eutectic Point.	MODERATE IMPORTANCE	2 Times Dec 2023, Jun 2023	Theory + Diagram

UNIT 4: SPECTROSCOPIC TECHNIQUES & CHROMATOGRAPHY

Topic 4.1: Spectroscopy (UV-Vis, IR, Rotational)

Beer-Lambert's Law: $A = \log(I_0 / I) = \epsilon \cdot c \cdot l$ where A is absorbance, ϵ is molar absorptivity, c is concentration, and l is path length.

Question Text	Classification	Occurrences / Years	Type
What is Spectroscopy ? State and explain Beer-Lambert's Law . Define related terms: Bathochromic, Hypsochromic, Hyperchromic shifts, and Chromophore.	MOST IMPORTANT	5 Times Dec 2023, Jun 2023, Jun 2022, Nov 2022, Jun 2020	Theory & Shifts
Describe IR (Infrared) / Vibrational Spectroscopy . State its selection rules, explain why some molecules are "IR active" while others are not, and describe types of molecular vibrations.	MOST IMPORTANT	6 Times Dec 2025, Jun 2024, Nov 2022, Dec 2020, Jun 2020, Jun 2023	Theory + Modes
Discuss Ultra-Violet (UV) / Electronic Spectroscopy principle, instrumentation, and applications. Detail the types of electronic transitions.	FREQUENTLY ASKED	4 Times Dec 2024, Dec 2023, Jun 2022, Nov 2022	Instrumentation
How can a combination of UV and IR spectroscopy be used to determine the molecular structure of an unknown organic compound?	RARE BUT POSSIBLE	1 Time Jun 2025	Application
Numerical Pattern (Electronic Spectroscopy): An unknown substance absorbs 40% of light at 435 nm in a 2 cm cell at a concentration of 20 g/L. What percentage of light is absorbed at 40 g/L?	RARE BUT POSSIBLE	1 Time Jun 2025	Numerical

Topic 4.2: Chromatography

Question Text	Classification	Occurrences / Years	Type
Explain the principle of Thin Layer Chromatography (TLC) . How is the Retention Factor (R_f value) calculated and what does it indicate?	MODERATE IMPORTANCE	1 Time Dec 2025	Theory

UNIT 5: PERIODIC PROPERTIES & BASIC CHEMICAL CONCEPTS

Topic 5.1: Periodic Trends (Groups & Periods)

Question Text	Classification	Occurrences / Years	Type
What is the Periodic Table? Write down and explain the trends of the following properties in a group and in a period with suitable reasons: (i) Electronegativity , (ii) Electron Affinity , (iii) Atomic & Ionic Size , (iv) Ionization Energy.	MOST IMPORTANT	6 Times Dec 2025, Jun 2025, Dec 2024, Jun 2024, Dec 2023, Dec 2020	Trends & Logic
Write precise technical notes on: (i) Effective Nuclear Charge (Z_{eff}) and Slater's rules, (ii) Polarizability (Fajan's Rule), (iii) Oxidation States variation, (iv) Aufbau principle.	MOST IMPORTANT	5 Times Dec 2025, Jun 2025, Dec 2024, Jun 2024, Jun 2023	Short Notes

FINAL ANALYSIS & STRATEGIC HIGHLIGHTS

RGPV Engineering Chemistry (BT-101) High-Scoring Matrix

OVERALL HIGH-SCORING & MOST REPETITIVE TOPICS

If you are short on time, mastering these 6 highly critical areas guarantees more than 80% of the exam weightage:

Core Pillar Topic	What to Focus On	Historical Weight
Water Softening Systems	Ion-exchange process schematic diagram, Zeolite reactions, and Lime-Soda numerical computations.	 Extremely High
Boiler Safeguards	Clear technical differentiation between Scales and Sludges; chemical context of Caustic Embrittlement.	 High Repetition
Lubricant Benchmarks	Mechanisms of Extreme Pressure lubrication and structural significance of Viscosity Index (VI).	 High Repetition
Polymer Syntheses	Step-by-step formation mechanism of Bakelite, PVC, and Free-radical chain-growth steps.	 Consistent
Corrosion Dynamics	Differential Aeration theory (pitting/waterline dynamics) and Cathodic safety mechanisms.	 Consistent
Periodic Architecture	Justifications for Electron Affinity and Electronegativity anomalies across periods.	 Predictable

Last-Minute Revision Checklist

- **Formulas to Memorize:** Hardness calculation factor (CaCO_3 equivalents), Viscosity Index formula ($VI = [(L - U) / (L - H)] \times 100$), and Gibbs Phase Rule equation ($F = C - P + 2$).
- **Diagrams to Practice:** (1) Deionizer column layout, (2) One-component water system phase graph showing triple point boundaries, (3) Pitting corrosion local cell formation.
- **Anomalies to remember:** Why Chlorine has higher electron affinity than Fluorine, and why Nitrogen has higher ionization potential than Oxygen.

*** End of Analysis Guide — Prepared for RGPV B.Tech First Year Engineering Students ***